

What is AI?

Learning Objectives

- Define artificial intelligence and distinguish between narrow and general AI
- Identify characteristics that define intelligent behavior in machines
- Evaluate current AI capabilities against human cognitive abilities

Engage (5-7 minutes)

Show a series of interactions: a chatbot conversation, a self-driving car navigating traffic, and a computer beating a world chess champion. Ask students: which of these require “intelligence”? Then pose the question: can a machine truly think, or does it only simulate thinking? This introduces the philosophical and technical foundations of AI.

Explore (10-12 minutes)

Students classify activities as human-only, machine-only, or both: recognizing faces, writing poetry, solving math problems, feeling emotions, playing strategy games, driving a car, translating languages. Group discussion: what separates true intelligence from pattern matching? How do we even define “intelligence”?

Explain (10-12 minutes)

Artificial Intelligence is the field of computer science dedicated to creating systems that perform tasks requiring human-like intelligence—perception, reasoning, learning, decision-making, and language understanding. Narrow AI (Weak AI) excels at specific tasks (image recognition, game playing) but cannot transfer skills to new domains. General AI (Strong AI) would match human cognitive flexibility across all domains—this remains theoretical. Current AI systems are all narrow, though increasingly capable within their domains.

Elaborate (8-10 minutes)

Trace the history of AI: Turing’s foundational question (1950), expert systems of the 1980s, the AI winters caused by overpromising, and the current deep learning revolution enabled by big data and GPUs. Examine the different approaches: symbolic AI (logic-based), machine learning (data-driven), and hybrid methods. Discuss how modern AI systems like large language models blur the line between narrow and general capabilities.

Evaluate (5-8 minutes)

Students write a reflective essay: “Can a machine be truly intelligent, or will it always be a sophisticated tool?” They must reference at least three AI systems and evaluate their capabilities honestly. Quiz: distinguish between narrow and general AI, identify examples of each, and explain why general AI remains elusive.